

## CT853 Algorithms and Logical Methods Assignment

**Due Date: 26/11/2021 @ 17:00**

For each of the questions below, you are required to provide:

1. A full code implementation in Java.
2. A short report in .pdf format that briefly explains how each algorithm works, and how your solutions were implemented, with reference to specific code snippets. Pseudocode is strongly encouraged for these explanations, but you may also use combinations of flowcharts, diagrams, plain text and code snippets also. This document should be no longer than 12 pages (max. 3 pages per question). A generic template is provided.

It is expected that additional independent research outside of the course notes will be required to complete this assignment. Take care to reference any websites/books/lecture notes that you make use of when writing your report. When you are finished, the .pdf document and your code should be uploaded in a .zipped folder to Blackboard. For questions 1 & 2 the objective of this assignment is about fully understanding these sorting algorithms and how they are implemented through your own explanation. We are not examining your ability to code and the sorting algorithm code in itself which is well known.

*Multiple submissions can be uploaded to Blackboard prior to the deadline but the last one will be the only one visible to our markers and the only one considered. This is an individual assignment and therefore the normal NUI Galway plagiarism rules apply regarding taking material from the web or from each other. In the event that you are informed or use ideas proposed in work that is published then normal referencing is expected. Plagiarised submissions will be penalised appropriately. Some students may be randomly selected to outline more details of their work in 1 to 1 meetings with the lecturer, and/or asked to resubmit their assignment.*

### Question 1 (10 Marks)

Write a Java program that can read in an array of numbers from the console, sort them using bubble sort, and output the sorted array to the console.

### Question 2 (10 Marks)

Write a Java program that can read in an array of numbers from the console, sort them using mergesort, and output the sorted array to the console. **Your implementation should be recursive.**

### Question 3 (10 Marks)

Write two Java programs (or alternatively one Java programme that includes the two alternative implementations) which can calculate the Fibonacci series to the value  $n$  where  $n$  is entered by the user via console.

- A. Code Fibonacci **without using recursion OR a function** (5 Marks)
- B. Code Fibonacci **using recursion**. (5 Marks)

The Fibonacci sequence is often referred to as the Fibonacci numbers. The first numbers in the series are 0, 1, 1, 2, 3, 5, 8..... Barring the first two terms in the sequence every other term is the sum of two previous terms. Take the following example  $8 = 3 + 5$  (This is the addition of 3 and 5).

### Question 4 (10 Marks)

Exponentiation, written in the form  $x^n$ , is a mathematical operation where the base  $x$  is multiplied by itself  $n$  times. Exponentiation by squaring is an algorithm that allows fast computation of this function. An outline of the exponentiation by squaring algorithm is provided below:

$$\text{expBySquaring}(x, n) = \begin{cases} \text{expBySquaring}(1/x, -n) & \text{if } n < 0 \\ 1 & \text{if } n = 0 \\ x & \text{if } n = 1 \\ \text{expBySquaring}(x * x, n/2) & \text{if } n \text{ is even} \\ x * \text{expBySquaring}(x * x, (n - 1)/2) & \text{if } n \text{ is odd} \end{cases}$$

Where  $\text{expBySquaring}(x, n)$  returns  $x$  to the power of  $n$ , and  $x$  and  $n$  are integers.

Write a Java program that reads in the two integers  $x$  and  $n$  from the console, computes the exponential using a **tail recursive implementation** of the exponentiation by squaring algorithm, and outputs the answer to the console.

END